

Conjecture 00000001543 is false:
 n collinear points on the irreducible curve $y = 0$ determine only
 $n - 1$ distances

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1 The conjecture

Conjecture 00000001543 (English text, verbatim): “Definition: Distinct distances on algebraic curves. Conjecture: Any n points on an irreducible real algebraic curve determine at least $c \cdot n^{4/3}$ distinct distances (the curve distance exponent).” The Chinese text says the same: any n points on any irreducible real algebraic curve determine at least $c \cdot n^{4/3}$ distinct distances.

Reading. Points lie in the Euclidean plane $\mathbb{R}^2$ with the Euclidean distance. An irreducible real algebraic curve is the zero set $Z(f) = \{p \in \mathbb{R}^2 : f(p) = 0\}$ of an irreducible polynomial $f \in \mathbb{R}[x, y]$; we additionally require $Z(f)$ to be infinite, which only shrinks the class (it excludes degenerate zero sets such as $Z(x^2 + y^2) = \{0\}$). The number of distinct distances of a finite set P is $\#\{|p - q| : p, q \in P, p \neq q\}$. The constant c is positive and independent of n . We refute the following readings, and claim no more:

(U) there is one $c > 0$ that works for every curve and every finite point set on it;

(C) every curve C has its own $c_C > 0$ and N_C such that every set of $n \geq N_C$ points on C determines at least $c_C n^{4/3}$ distances.

(C) is the weakest of the natural quantifier orders (constant depending on the curve, bound only for large n), so (U) and every intermediate order (constant depending on the degree, bound for all n) are refuted too.

Not refuted. The statement as written does not exclude lines or circles. The known theorem of Pach and de Zeeuw (Section 4) does exclude them, and the reading “irreducible curves other than lines and circles” is a true theorem; we do not refute it.

2 The counterexample

Lemma 1. *The x -axis $L = \{(x, y) : y = 0\} = Z(y)$ is an irreducible real algebraic curve.*

Proof. The variable y is a prime, hence irreducible, element of $\mathbb{R}[x, y]$, and $Z(y)$ contains the infinitely many points $(t, 0)$, $t \in \mathbb{R}$. □

Lemma 2. *Let $P_n = \{(k, 0) : k = 0, 1, \dots, n - 1\} \subset L$. Then $|P_n| = n$ and P_n determines at most $n - 1$ distinct distances.*

Proof. $|(a, 0) - (b, 0)| = |a - b|$, and for $a \neq b$ in $\{0, \dots, n-1\}$ this is an integer in $\{1, \dots, n-1\}$. $\square$

Lemma 3. *For every $c > 0$ and every integer $n \geq \lceil c^{-3} \rceil + 1$ we have $n - 1 < cn^{4/3}$.*

Proof. $n > c^{-3}$ gives $(cn^{1/3})^3 = c^3n > 1$, so $cn^{1/3} > 1$ and $n - 1 < n < n \cdot cn^{1/3} = cn^{4/3}$. $\square$

Theorem 4. *For every $c > 0$ there is N such that for every $n \geq N$ some n points of the irreducible curve L determine fewer than $cn^{4/3}$ distinct distances. Consequently readings (U) and (C) are false.*

Proof. Take $N = \lceil c^{-3} \rceil + 1$ and P_n ; combine Lemmas 2 and 3. For (U): apply this to the given c and L . For (C): given c_L, N_L , apply it with $c = c_L$ and $n = \max(N_L, N)$. $\square$

The same failure occurs on circles (a regular n -gon determines $\lfloor n/2 \rfloor$ distances); we only use the line, and the circle remark is not formalized.

3 Lean formalization

File `lean/Conjecture1543/Basic.lean` (173 lines, only `import Mathlib`).

- `Plane := EuclideanSpace R (Fin 2)` (Euclidean metric; coordinates `p 0, p 1`).
- `zeroSet f`: $\{p : \text{MvPolynomial.eval}(i \mapsto p i) f = 0\}$ for `f : MvPolynomial (Fin 2) R`.
- `IsIrreducibleRealCurve C`: $\exists f, \text{Irreducible } f \wedge C = \text{zeroSet } f \wedge C.\text{Infinite}$.
- `distances P`: the image of $(p, q) \mapsto \text{dist } p q$ over `P.offDiag` (ordered pairs of distinct points); its `card` is the number of distinct distances.
- `xAxis := zeroSet (X 1)`; `xAxis_isIrreducibleRealCurve` (Lemma 1, via `Mathlib's MvPolynomial.X_prime`).
- `collinear n, collinear_card, collinear_subset, distances_collinear_card_le` (`card` $\leq n - 1$; Lemma 2).
- `eventually_lt` (Lemma 3; real power `Real.rpow`).
- `xAxis_violates` (Theorem 4), `not_uniform` (reading U), `not_per_curve_eventually` (reading C).

Axiom audit: `#print axioms` for `xAxis_violates`, `not_uniform` and `not_per_curve_eventually` each reports `[propxt, Classical.choice, Quot.sound]`. No `sorry`, no `native_decide`, no added axioms.

4 Relation to the known theorem and to the earlier submission

Pach and de Zeeuw, *Distinct distances on algebraic curves in the plane*, <https://arxiv.org/abs/1308.0177>, abstract: “We prove that the number of distinct distances determined by P is at least $c_d n^{4/3}$, unless C contains a line or a circle.” The conjecture’s text drops the exception “unless C contains a line or a circle”, and the exception is necessary: Theorem 4 is exactly the line case.

Relation to the earlier submission. PR #221 (by orionsheep) used the same counterexample, but its Lean theorem `no_constant` was pure natural-number arithmetic. The review rejected it because “the actual counterexample (collinear points determine $n - 1$ distances) is never formalized — curves, points, and distances appear only in prose/Python.” This submission formalizes the objects themselves: the Euclidean plane, irreducible polynomials in `MvPolynomial (Fin 2) ℝ` and their zero sets, finite point sets on the curve, and the distinct-distance count as the cardinality of the image of the distance map. It proves the violation for every $c > 0$ and all large n , under both a uniform and a curve-dependent constant.